ITM SKILLS UNIVERSITY

Department of Computer Science & Engineering

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

PETPAL

Case Study No. 64

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Enterprise Business Problem, As-Is/To-Be Analysis, Stakeholder Matrix & Scope Management

SUBMITTED BY

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Project: PetPal — Appointment Booking and Records App for a Veterinary Clinic Chain
Case Study: No. 64
Subject: Software Engineering & Project Management

SECTION 1 & 2

1. Business Problem & Existing Workflow (As-Is)

Across the five (5) Bengaluru veterinary clinics (*Indiranagar, Koramangala, Whitefield, Jayanagar, Bellandur*), appointment booking is conducted purely via manual telephone calls. Over 300 daily inbound calls overwhelm receptionists, resulting in a 22% call abandonment rate, frequent double-booking errors, and a high no-show rate of approximately 18%. Furthermore, pet vaccination cards are paper-based and frequently lost, causing pet parents to miss crucial annual booster immunizations.

2. Proposed Business Solution (To-Be)

PetPal introduces a centralized 24/7 web booking portal and clinic management dashboard:

- **24/7 Self-Service Booking:** Pet owners discover real-time 30-min available slots across all 5 branches.
- **Digital Pet Health Profiles:** Central registry of pet breed, age, weight, and microchip tagging.
- **Automated Immunization Recall:** Rule-based engine dispatches reminders 14d, 3d, and 1d prior to booster dates.
- **Online Pre-Payment Deposits:** Reduces no-show rates from 18% to under 5%.
- **Live Clinic Reception Queue:** Front-desk staff manage patient arrivals with one-click check-ins.

SECTION 3 & 4

3. Business Objectives & 4. Stakeholder Matrix

TABLE 4.1: STAKEHOLDER ANALYSIS MATRIX

STAKEHOLDER	ROLE	INTEREST	INFLUENCE	RESPONSIBILITIES	SYSTEM INTERACTION
Pet Owner	Customer	High	High	Book visits, maintain pet profiles, pay deposits.	Web Portal (Screens 01-17)
Chief Veterinarian	Medical Director	High	High	**Enforce zero clinical notes duplication**.	Governance audits
Managing Partner	Executive Sponsor	High	High	Approve budgets, monitor chain utilization ROI.	Executive Reports (Screen 26)
Veterinarian	Clinical Care	Medium	High	Conduct consultations, review daily rosters.	Staff Dashboard (Screen 19)
Receptionist	Operations	High	Medium	Check in arriving pets, manage live queue.	Live Queue (Screen 20)
Existing Clinic EMR	System of Record	Medium	High	Maintain clinical notes; sync schedule metadata.	Stateless REST APIs
Payment Gateway	External FinTech	Low	Medium	Capture UPI/Card transactions securely.	HTTPS Webhooks
Notification Svc	Messaging Infrastructure	Low	Medium	Deliver SMS/Email alerts within SLA.	Message Queue API

SECTION 5 & 6

5. Business Scope & 6. Change Request CR-01 (Pet-Food Store)

SCOPE CREEP GOVERNANCE & EVALUATION

The Managing Partner requested adding an online e-commerce pet-food store (+55 person-days, +₹1,65,000). Although the revised cost of ₹4,26,000 remains below the ₹8,00,000 budget cap, this is **severe scope creep**. It extends delivery by 11 weeks, introduces retail inventory liabilities, and dilutes focus from clinical booking.

TABLE 6.1: FORMAL SCOPE CHANGE REQUEST CR-01

CHANGE REQUEST ID	**CR-01**	SUBMISSION DATE	2026-10-02
TITLE	Addition of Online E-Commerce Pet-Food & Accessory Store		
REQUESTED BY	Managing Partner	EVALUATION STATUS	DEFERRED TO PHASE 2
EFFORT & COST IMPACT	+55 person-days (+₹1,65,000) | Revised Total: 142 person-days (₹4,26,000 vs ₹8,00,000 budget)		
CCB FINAL DECISION	**FORMALLY DEFERRED TO PHASE 2** to protect the 85-day critical path and core clinical launch.		